

Lesson 8 $\mathfrak{sl}(2, F)$ -module

Monday, March 30, 2026

20:09

Continued proof of Weyl's Thm

$0 \neq W < V$ is an arbitrary submodule of V .

We are looking for a linear transformation $P: V \rightarrow W$ such that $\forall w \in W, P(w) = w \Rightarrow P^2 = P$, P is a projection $\Rightarrow V = W \oplus \ker P$

We need P to be a L -module homomorphism $\Leftrightarrow P$ commutes with L

$\Leftrightarrow L$ acts on P trivially

(Recall that $\forall a \in L, P \in \text{Hom}_F(V, W), v \in V$,

$$[a, P](v) = a \cdot P(v) - P(a \cdot v)$$

Let $T = \{ \varphi: V \rightarrow W \mid \varphi|_W = \alpha \text{Id for some } \alpha \in F \}$

T is in fact an L -submodule: $\forall \varphi \in T, a \in L, w \in W$

$$[a, \varphi](w) = a \cdot \varphi(w) - \varphi(a \cdot w) = 0.$$

Setting $T_0 = \{ \varphi: V \rightarrow W \mid \varphi|_W = 0 \} \subseteq T$, we have in fact show $[L, T] \subseteq T_0$. In particular, T_0 is an L -submodule of T with codimension 1. By previous argument, $T = T_0 \oplus F\varphi$ for some 1-dimensional submodule $F\varphi$ and L acts trivially on φ . WLOG, assume $\varphi|_W = \text{Id}_W$, then $V = W \oplus \ker \varphi$.

Rmk Recall Maschke's thm

Let G be a finite group, K is a field whose characteristic does not divide $|G|$, then every $K(G)$ -module is completely reducible.

pf: We prove by showing any submodule W of a $K(G)$ -module V is a direct summand.

We want a projection $P: V \rightarrow W$ s.t. P commutes with G .

If not, we modify P : consider $P' = \frac{1}{|G|} \sum_{g \in G} gPg^{-1}$ $\square$

For Weyl's thm, the original proof by Weyl uses this method on (compact) Lie groups (if not, consider compact Lie subgroups)

sl₂ - module

sl₂ basis: e, f, h (sl₂-triple)

$$[e, f] = h, [h, e] = 2e, [h, f] = -2f$$

$$e \leftrightarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, f \leftrightarrow \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, h \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Let V be an irreducible module over sl₂.

Write $V = \bigoplus V_\alpha$ where V_α are root spaces of $h: V \rightarrow V$

Let v be an α eigenvector of h , say $h v = \alpha v$.

$$\text{Then } h(e v) = [h, e] v + e(h v)$$

$$= 2e v + \alpha e v$$

$$= (\alpha + 2) e v$$

$$h(f v) = [h, f] v + f(h v)$$

$$= -2f v + \alpha f v$$

$$= (\alpha - 2) f v.$$

$$\text{i.e. } v, e v, e^2 v, \dots \quad \exists 0 \neq v \text{ s.t. } h v = \lambda v, e v = 0.$$

$$\alpha, \alpha + 2, \alpha + 4, \dots$$

$$\text{On the other hand, } v, f v, f^2 v, \dots, f^m v \neq 0, f^{m+1} v = 0$$
$$\lambda \quad \lambda - 2 \quad \lambda - 4 \quad \lambda - 2m$$

Claim: $v, f v, \dots, f^m v$ is a basis of V

1° $f^i v$ are linearly independent since they have different eigenvalues

2° consider $e(f^i v)$:

$$e f^i v = e f f^{i-1} v$$

$$= f e f^{i-1} v + (\lambda - 2(i-1)) f^{i-1} v$$

$$= f^2 e f^{i-2} v + (\lambda - 2(i-2)) f^{i-1} v + (\lambda - 2(i-1)) f^{i-1} v$$

$$\vdots \quad [(\lambda - 2(i-1)) + (\lambda - 2(i-2)) + \dots + (\lambda - 2(0))] f^{i-1} v$$

$$= (i\lambda - (i-1)i) f^{i-1} v$$

$$= i(\lambda - i + 1) f^{i-1} v$$

Now, applying to $i = m+1$,

Now, applying to $i=m+1$,

$$\text{LHS} = 0 = (m+1)(\lambda - m) f^m v$$

Since $f^m v \neq 0$, conclude that $\lambda = m$.

Thus $v, f v, \dots, f^m v$
 $m, m-2, \dots, -2m$

This is:

Thm reps of sl_2

① In every dimension ≥ 1 there exists one irreducible module V of that $\dim m+1, m \geq 0$, $\exists$ a highest weight elt v , $h v = m v$, $e v = 0$.
 $\{v, f v, \dots, f^m v\}$ is a basis of V .

$m, m-2, \dots, -m$

② Let V be a f.d. sl_2 -module, then any eigenvalue of h in V is an integer.

③ Let V be a f.d. sl_2 -module. Let w be eigenvector that belongs to an eigenvalue α , then $\alpha = q - p$ where p, q satisfy
 $f^{q+1} w = 0, f^q w \neq 0, \dots, f w, w, e w, e^2 w, \dots, e^p w \neq 0, e^{p+1} w = 0$.

pf of ③:

First suppose $w = f^i v$ for some highest weight elt v and for some i .
i.e. $w \in \{v, f v, \dots, f^m v\}$. Then $\alpha = m - 2i, q = m - i, p = i$
 $\Rightarrow \alpha = q - p$.

Now for any $w = f^{i_1} v_1 + \dots + f^{i_k} v_k$ for some highest weight v_j and for some i_j where each $f^{i_j} v_j$ has the same eigenvalue. Using the same notation, we then have $\alpha_j = q_j - p_j = q_i - p_i = \alpha_i$

Then for w , $p = \max(p_1, \dots, p_k), q = \max(q_1, \dots, q_k)$ and $\alpha = q - p$ as desired.

Rmk existence for rep of any dimension $m+1, m \geq 0$

Consider $F[x, y]$.

Set $e = y \frac{\partial}{\partial x}, f = x \frac{\partial}{\partial y}, h = [e, f] = y \frac{\partial}{\partial y} - x \frac{\partial}{\partial x}$

Set $e = y \frac{\partial}{\partial x}$, $f = x \frac{\partial}{\partial y}$, $h = [e, f] = y \frac{\partial}{\partial y} - x \frac{\partial}{\partial x}$
 $V = \text{span} \{ x^i y^j \mid i+j=m \} \Rightarrow \dim V = m+1$
 $\Rightarrow$ The highest weight elt is $v = y^m$